

ORF307: Optimization

Fall Semester 2025 – 1st Midterm Exam

October 9, 10:40 am – 12:00 pm

Exam Rules

1. Please write out and sign the following pledge on the solution booklet: “I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam.”
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. There are 3 problems worth a total of 100 points.
6. At the end of the exam, please turn in both your solution booklet and the exam questions.

Problem 1 (30 points). *Modeling Humanitarian Support under Constraints*

Following a major earthquake, a humanitarian organization must procure three types of supplies - food (x_1), water (x_2), and medical kits (x_3) - to support four affected regions: R_1, R_2, R_3 and R_4 . The decision variables $x = (x_1, x_2, x_3)^T \in \mathbb{R}^3$ represents the quantities of each supply to be purchased. The organization uses a known matrix $A \in \mathbb{R}^{4 \times 3}$ that maps the procurement vector x to the utility delivered to each region: $\hat{b} = Ax$, where $\hat{b} \in \mathbb{R}^4$ and $\hat{b}_i$ represents the utility received by region R_i , for $i = 1, 2, 3, 4$. The governor of each region has requested a specific utility level to be delivered to her/his region. The organization captures these requests in the demand utility vector $b \in \mathbb{R}^4$.

The organization's main goal is to match, as closely as possible, the delivered utility to the requested utility, subject to the following constraints:

- **Budget constraint:** Given that each unit of supply i costs c_i dollars, the total procurement cost must not exceed the organization's annual budget β .
- **Policy constraint (Region Priority):** Region R_2 is identified as especially vulnerable and must receive adequate support. For this reason, the utility delivered to R_2 must be at least 90% of the utility level requested by its governor.

Note: The utility delivered to each region is determined by a weighted combination of the supplies procured. Specifically, the utility delivered to region R_i corresponds to the inner product of the i -th row of matrix A with the procurement vector x .

- **Minimum Total Utility:** The sum of the utilities delivered to all four regions must be greater than or equal to a known critical threshold, U_{min} .
- **Nonnegativity:** Negative procurement is not allowed.

Your tasks for this problem are:

- (10 points) Formulate a constrained optimization problem that minimizes the L_1 -norm of the deviations between delivered and requested utility.
- (5 points) Reformulate the optimization problem in Part a. as an LP problem.
- (10 points) Formulate a constrained optimization problem that minimizes the L_∞ -norm of the deviations between delivered and requested utility (the constraints are the same as in Part a.)
- (5 points) Reformulate the optimization problem in Part c. as an LP problem.

Problem 2 (35 points). *Scheduling with Managerial Constraints*

Alice has just started her summer internship as an Optimization Specialist in a company that manufactures two products: P1 and P2. One unit of each product needs 1 hour to be produced. The company earns \$1 and \$1.5 of profit when it sells one unit of products P1 and P2, respectively. John, the manager of the company, gave Alice the task of optimizing the weekly production scheduling of these two products. He told her that she has up to 100 hours of machine time available every week and this time should be allocated diligently between the two products so that the total profit of the company is maximized.

John told Alice that due to some contractual obligations and safety requirements, she must allocate at least 20% of the time to product P1 and no more than 40% of the time to product type P2. Another additional requirement that John emphasized was that the time allocated to either product should not exceed the time allocated to the other by more than 30% of the total available time.

- a. (10 points) Formulate a linear programming problem to maximize the total profit of the company subject to the constraints that the manager specified.
- b. (10 points) Plot the feasible region defined by the constraints. Compute the coordinates of all the vertices of the feasible region.
- c. (10 points) Draw three contour lines of the objective function that pass through the feasible region. Identify the optimal solution and briefly justify it graphically. Compute also the optimal objective value.
- d. (5 points) After plotting the feasible region can you identify any constraints that may be redundant (a constraint is redundant if it does not affect the feasible region at all)? Briefly explain your reasoning.

Problem 3 (35 points). *Regularized Weighted Least Squares.*

Consider the least squares optimization problem

$$\min_{x \in \mathbb{R}^n} \quad \mu \|Ax - b\|^2 + \lambda \|x\|^2 + \|Cx - d\|^2 \quad (1)$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{p \times n}$, $d \in \mathbb{R}^p$.

- a. (10 points) Define a new matrix $\tilde{A}$ and a new vector $\tilde{b}$ such that the optimization problem (1) can be written in the standard least squares form:

$$\min_{x \in \mathbb{R}^n} \quad \|\tilde{A}x - \tilde{b}\|^2 \quad (2)$$

Also, state the exact dimensions of the matrix $\tilde{A}$ and vector $\tilde{b}$.

- b. (10 points) Derive the normal equations for the least squares problem in standard form defined by (2) and next express them in terms of the matrices and vectors (A, b, C, d) and parameters μ, λ of the original least squares problem defined by (1).
- c. (15 points) Now suppose we add the equality constraints $Fx = h$ to define the constrained least squares problem:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \|\tilde{A}x - \tilde{b}\|^2 \\ \text{s.t.} \quad & Fx = h \end{aligned} \quad (3)$$

where $F \in \mathbb{R}^{q \times n}$ and $h \in \mathbb{R}^q$. Define the Lagrangian function of this problem and write the KKT conditions. From the KKT conditions derive the optimal solution $\tilde{x}^*$ of problem (3).

Note: All norms in the problem are L_2 (Euclidean) norms.